

Practice Quiz 04A — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 04

Set identities by logical laws; unions of families and power sets.

1. (a) Let A, B be sets. Prove $A \cup (B \setminus A) = A \cup B$ by a chain of equivalences beginning with membership in the left side. Name the definition or logical law used at each step.

Solution Let x be arbitrary, and abbreviate the statements $x \in A$ and $x \in B$ by P and Q , respectively. The definitions of union and difference give

$$x \in A \cup (B \setminus A) \iff P \vee (Q \wedge \neg P).$$

Now apply the logical laws:

$$\begin{aligned} P \vee (Q \wedge \neg P) &\iff (P \vee Q) \wedge (P \vee \neg P) && \text{(distributivity)} \\ &\iff (P \vee Q) \wedge \text{true} && \text{(excluded middle)} \\ &\iff P \vee Q && \text{(identity law).} \end{aligned}$$

By the definition of union, $P \vee Q$ is equivalent to $x \in A \cup B$. Thus the two sets have exactly the same elements and are equal. ■

- (b) Let $\mathcal{F} = \{\{2\}, \{2, 5\}\}$. Compute $\bigcup \mathcal{F}$ and $\mathcal{P}(\bigcup \mathcal{F})$.

Solution The union is $\bigcup \mathcal{F} = \{2, 5\}$. Its subsets give

$$\mathcal{P}(\bigcup \mathcal{F}) = \{\emptyset, \{2\}, \{5\}, \{2, 5\}\}. \quad \blacksquare$$